An AI-Powered Platform to Bridge the Gap Between Sri Lankan Citizens and Government Services

Project Proposal Report

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Group Number - 25-26J-153
Date - 08/31/2025

B.Sc. (Hons) Degree in Information Technology Specializing in Information System Engineering

Department of Information Technology
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Machine Learning–Based Resolution Time Prediction and Hybrid Prioritization Module for GovConnect LK

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DECLARATION

I hereby declare that the research proposal is my original work and has been prepared by me. The information and data presented in this report are true and accurate to the best of my knowledge. All sources of information and literature used in this report have been properly cited and acknowledged.

I further declare that this proposal has not been submitted, in whole or in part, for any other degree or qualification at this or any other institution, and it complies with the ethical standards of research as outlined by Sri Lanka Institute of Information Technology.

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The supervisor/s should certify the proposal report with the following declaration. The above candidates are carrying out research for the undergraduate Dissertation under my supervision.

Signature of supervisor:- Date :- 27/08/25

Signature of co-supervisor:- Date :- 28/08/2025

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ABSTRACT

Sri Lanka's public complaint management process is often slow, manual and inefficient which leads to delays, misclassification and inattention to prioritization of issues. In addition, the complainants often use informal Sinhala-English code-mixed text, further defying automated systems and marginalizing rural communities with lower levels of digital literacy. To bridge these gaps, this project aims to develop GovConnect LK, an AI-based complaint management system which utilizes Natural Language Processing (NLP) and Machine Learning (ML) to enhance efficiency, transparency and responsiveness in government service delivery. The NLP engine that is trained on Sinhala-English and code-mixed datasets, automatically categorizes the complaints into categories and detects urgency. A companion ML model is used to estimate the expected resolution time from historical or simulated data. These outputs are then paired with community voting signals and generate a dynamic priority score for smarter complaint handling. Complaints and categorization are stored in a centralized database and visualized in a real-time officer dashboard to allow government officials to track, prioritize, and effectively resolve grievances. The originality of this research consists in the possibility to treat informal multilingual inputs, in the hybrid prioritization algorithm, and in the integration of citizen feedback into the decision-making process. In conclusion, GovConnect LK holds the promise of transforming the way public grievance is managed, making it speedier, fairer, and more inclusive in Sri Lanka, while serving as a model for AI-driven governance in the developing world.

ACKNOWLEDGEMENT

We wish to thank most heartily all those individuals and institutions that gave us advice, help, and encouragement during the successful accomplishment of this research project.

First of all, we wish to thank our supervisor Buddima Aththanayaka for her invaluable guidance, constructive criticism, and constant encouragement, which have greatly shaped the vision and the direction of this project. Their wise counsel and knowledge were responsible to a significant extent for the quality and depth of our research. We are also indebted to our co-supervisor, Narmada Gamage, for her constructive comments, technical expertise and continuous encouragement to push us in overcoming the difficulties and in improving the methodology.

We also wish to express our deepest gratitude to the Department of Information Technology of SLIIT for offering the academic platform, resources and infrastructure needed to carry out this research. The department's support created a productive environment for experimentation, learning and collaboration.

Our thanks also go out to our colleagues and peers who shared knowledge, gave feedback and motivation throughout the project. Their constructive criticism and support was immensely appreciated and helped in the successful completion of this work.

Lastly, we want to thank our families and friends for their patience, understanding and moral support on this journey. Their continuous support and faith in us, motivated us to keep going and achieve this research.
To you all, we are very much in debt; it is without your direction, support, and encouragement that this success would not have been realized

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LIST OF ABBREVIATIONS

Abbreviation	Full Form
AI	Artificial Intelligence
API	Application Programming Interface
AWS	Amazon Web Services
DB	Database
EC2	Elastic Compute Cloud
GPU	Graphics Processing Unit
HTTPS	Hypertext Transfer Protocol Secure
JSON	JavaScript Object Notation
JWT	JSON Web Token
MAE	Mean Absolute Error
ML	Machine Learning
MongoDB	Mongo Database (NoSQL)
NLP	Natural Language Processing
R^2	Coefficient of Determination
REST	Representational State Transfer
RMSE	Root Mean Square Error
SaaS	Software as a Service
S3	Amazon Simple Storage Service
SMS	Short Message Service
UI	User Interface
UX	User Experience
VS Code	Visual Studio Code
WCAG	Web Content Accessibility Guidelines
XGBoost	Extreme Gradient Boosting

1.INTRODUCTION

1.1. Background of the Research

In Sri Lanka, the current public complaint process continues to be manual, paper-based and inefficient. Citizenstypically raise their grievances by writing letters, calling a hotline, or using informal channels that are then manually channelled by government officials. This process leads to delays, high misclassification probability and lack of a structured approach to assess urgency. As a consequence, problems of public safety and minor complaints are given the same priority, which diminishes citizen trust and confidence in government service delivery.

Moreover, the problems are even more serious in rural areas where a very large percentage of citizens struggle with language barriers and have little digital literacy. Complaints are frequently written in informal Sinhala, English or code-mixed text, which most automated systems do not correctly interpret. Further, officials currently have no analytical tools to determine how long a complaint may take to be resolved, which means prioritization is solely based on subjective judgment. This results in discrepancies, inefficiencies, and an inefficient distribution of resources.

In order to tackle these challenges, the GovConnect framework has been proposed as an intelligent ICT solution for the public grievance handling. While complaint classification and urgency detection can be assisted by Natural Language Processing (NLP) techniques, an important gap remains in forecasting how long a complaint will take to be resolved and in the design of a fair prioritization scheme. By applying Machine Learning (ML) regression models trained on historical and simulated data, accuracy of the resolution times can be improved.

More specifically, the research presented here aims at designing a hybrid prioritization model that combines three criteria: urgency, expected resolution time and citizen voting signals. The model will be implemented into an interactive officer dashboard, that will allow filtration, sorting, and decision-making in real-time. Such an approach will help ensure that critical high-impact complaints are dealt with effectively while ensuring transparency, accountability and citizen participation. Finally, this work illustrates the use of Information Systems Engineering principles to design a data-driven, citizen-oriented, and technologically innovative grievance management system for Sri Lanka.

1.2Literature Survey

Artificial Intelligence (AI) has recently become more popular in the domain of public grievance management, but most of the current approaches focus on classification of the grievance complaints and routing them, and do not tackle the problem of predicting time to resolution and prioritizing them fairly. E-government complaint systems have historically concentrated on digitization of submission and tracking activities where complaints are entered and directed to appropriate departments but with no analytical support of urgency or time to completion [1]. Studies on complaint resolution in developing countries have identified ineffectivities in manual triage, and reported that officials use a great amount of personal judgment, resulting in delays and inconsistent outcomes [2], [5].

Machine Learning (ML) regression models (such as Random Forests and XGBoost) have been successfully used in services delivery forecasting and resource scheduling contexts, showing promise for predicting task completion times from history [3], [6]. However, these studies focus much on the context of business or logistics and have not been generalized to public service complaints. Some hybrid prioritization approaches have been

tried in the healthcare triaging and emergency response systems, in which urgency and estimated resolution time are fused to allocate resources [4]. Although these approaches are found to be highly applicable to public complaint management, their applicability to local governance has not been studied much.

Citizen engagement mechanisms have also been researched in smart governance systems, especially through techniques based on voting or crowdsourcing that allow the community's priorities to be identified [5], [6]. The participatory governance literature highlights that the incorporation of citizen signals can help improve fairness and responsiveness of governance, but most systems fail to have an integrated framework that fuses citizen voting with predictive analytics. Previous work has suggested dashboards that can be used by government officials to increase transparency and visualization, but many of these dashboards are static and lack real-time feedback loops to update the status of complaints or facilitate interactive decision making [7], [8].

1.3 Research Gap

The existing procedures employed in managing and ranking the complaints received by the people in Sri Lanka are largely manual, in which government officers go through complaints and refer them to departments, according to personal opinion. Such manual method adds a lot of subjectivity, which commonly leads to mis-classification of complaints, slowness in resolution and disproportionate prioritization. As a result, the urgent matters, like the issues related to the safety of the population might get the same level of attention as the less important complaints related to services, causing the dissatisfaction of the citizens and damaging the trust in governmental services.

One particular weakness is that the current e-government complaint management systems are based on first come, first served or primitive queue based model. Such systems do not consider the urgency, severity or the impact of complaints on the society. An example is that a complaint of a broken water supply line with hundreds of residents impacted may come behind an issue that has a low impact, such as a minor road repair request, and be served because it was lodged later. Such a model is not representative of the real urgency or possible impacts of complaints, which postpones the solving of high-priority issues and improves the general performance of the delivery of public services.

The final limitation is another critical one because the existing complaint management practices lack predictive analytics. Government officials have no tools to estimate how long complaints will be resolved, by historical information, by the type or by the severity. In the absence of predictive insights, decisions are based on experience as opposed to data-based judgments. This slows down the effectiveness of resource allocation, eliminates the possibility of proactive complaint management, adds to longer resolution times.

Also, there is low citizen participation in the prioritization. Although the complaints are received electronically, citizens are not excessively involved in the decision on what issues should be resolved the first. Such absence of community input may lead to discontent, especially when grievances of many persons are ignored because of administrative custom instead of the pressing demands of reality.

All these gaps demonstrate the necessity of a superior system that can estimate the urgency of complaints automatically using machine learning to predict the time of its resolution, and introduce the element of citizen voting to pay attention to the interests of the community. The proposed system would allow managing complaints more efficiently and fairly and provide an insight into them through a real-time officer dashboard combined with these features into a hybrid prioritization algorithm. This would not only make the operations smoother, but also increase transparency, active governance and overall boost citizen confidence in the services of the state.

Also, the combination of historical complaint data and machine learning models may help to understand trends and recurring patterns and allow government agencies to recognize systemic problems and allocate resources in a more intelligent way. Multi-level escalation may also be facilitated in the proposed system since complaints with high impact are automatically set to urgent attention. These improvements would over time lead to a more intelligent, responsive public service ecosystem in Sri Lanka.

Research Area / Feature	**Research 1**	**Research 2**	**Research 3**	**Research 4**	**Research 5**	**Proposed Solution**
Manual complaint handling and prioritization	Yes	Yes	Yes	Yes	Yes	No
Predictive complaint resolution time	No	No	No	No	No	Yes
Urgency-based prioritization	No	No	No	No	No	Yes
Citizen voting integration	No	No	No	No	No	Yes
Hybrid prioritization algorithm	No	No	No	No	No	Yes
Real-time officer dashboard	No	No	No	No	No	Yes

Table 1:Research Gap

1.4 Research Problem

Current methods of managing and prioritizing public complaints in Sri Lanka are limited in accuracy and effectiveness due to their reliance on manual review and subjective judgment by government officers.
The first question that arises is how to accurately predict the resolution time of complaints. Without a reliable prediction model, officers cannot estimate how long a complaint may take to resolve, which makes prioritization arbitrary and inconsistent. Understanding the factors that influence resolution time—such as complaint type, urgency, and responsible department—is crucial for developing a method that can move beyond guesswork and provide data-driven insights.
The second question is how to design a prioritization method that reflects not only urgency but also predicted resolution time and citizen participation. Existing systems often prioritize complaints on a *first come, first served* basis, which means urgent or high-impact complaints can be delayed. Identifying the right hybrid approach—one that combines urgency detection (from NLP outputs), machine learning–based resolution time prediction, and citizen voting signals—can ensure complaints are ranked in a fair and impactful way.

Another question is how to implement this prioritization framework into a system that supports real-time decision-making. This involves building the backend infrastructure to manage complaints, generate priority scores, and route requests efficiently, while also providing an officer dashboard with interactive features such as filtering, sorting, and visual analytics. The system must be both technically robust and user-friendly to ensure adoption by government officers.

Finally, there is the question of how to integrate feedback loops that update complaint statuses and notify citizens in real time. Such feedback mechanisms are essential for transparency and for maintaining citizen trust, as they allow stakeholders to monitor progress and hold authorities accountable. By addressing these challenges, this research seeks to deliver a hybrid prioritization system that transforms complaint management into a

smarter, fairer, and more efficient process, ultimately improving public service delivery

2.OBJECTIVES

2.1 Main Objective

The primary goal of such a hybrid prioritization system is to correctly forecast the time of complaint resolution and, based on the forecasts and the urgency and citizen feedback, establish a reasonable system of scheduling and decision making that minimises critical complaints to receive prompt treatment without undermining transparency and citizen confidence.

2.2 Sub Objectives

The 4 major sub objectives are as follows:

- **To collect or simulate a dataset** of complaints containing resolution time, department, urgency, and type to form the foundation for predictive modeling.

- **To train a regression model** (e.g., XGBoost or Random Forest) that is able to estimate resolution time based on the type of complaint with historical or simulated data.

- **To design a hybrid prioritization algorithm** combining urgency (as a result of NLP analysis), ML-predicted resolution time, and community voting indicators to form a dynamic priority score.

- **To implement an intelligent officer dashboard** (using Node.js, Express, and React.js) to allow storing complaints, generating priority scores, filtering, sorting, analytics visualization, and real-time feedback loops to update complaint statuses and inform citizens.

3.METHODOLOGY

- **Data Collection**

Reliable sources like Gamdata, Rajya Thorrathutu Kendraya and other government bodies will be used to gather a dataset of complaints by the people. The data will cover such aspects as the time of resolving the complaints, departments, urgency levels, and the type of complaints. In case of a lack of historical data, simulated datasets will be created to give a full coverage. The data obtained will undergo preprocessing to deal with missing values, normalize features and to prepare it to undergo machine learning analysis.

- **Resolution Time Prediction Model**

Based on the data obtained, a regression model (e.g., XGBoost or Random Forest) will be trained on how to predict the time to resolve various kinds of complaints. The model will apply characteristics like the type of complaint, department and the urgency. The performance will be measured using such metrics as MAE, RMSE, and R 2, which will provide accurate predictions.

- **Hybrid Prioritization Algorithm**

There will be a computed hybrid priority score (the combination of three factors):

1. Urgency, obtained with the help of the NLP analysis of the text of complaint
   .
2. Anticipated resolution time which the regression model estimates.

3. Citizen feedback and priorities in the form of community voting

The hybrid algorithm will also dynamically compute a priority score in each complaint, so that the most important problems can be addressed in time, but also representing the interests of the community.

- **Backend System Implementation**

The backend is going to be developed on the basis of Node.js and Express, which will store complaints, process API requests, form priority scores, and ensure data consistency in real-time. Feedback loops will be implemented to enable updating of complaint statuses and automatic notifications to the users.

- **Officer Dashboard**

React.js will be used to create an interactive officer dashboard that will have functionalities like sorting and filtering of complaints and analytic visualization. Officers will monitor the real-time priority scores, track the progress in resolving complaints and make data-driven decisions in an effective manner.

- **Evaluation and Novelty**

The system shall be measured against accuracy in prediction, enhancement of efficiency, fairness in priorities as well as satisfaction of citizens. The originality of the research is that the hybrid prioritization algorithm is unique and combines NLP-based urgency detection, machine learning-predicted resolution time, and citizen voting to realize the idea of smarter, faster, and fairer complaint resolution in the real world governance system.

3.1.System Architecture Diagram

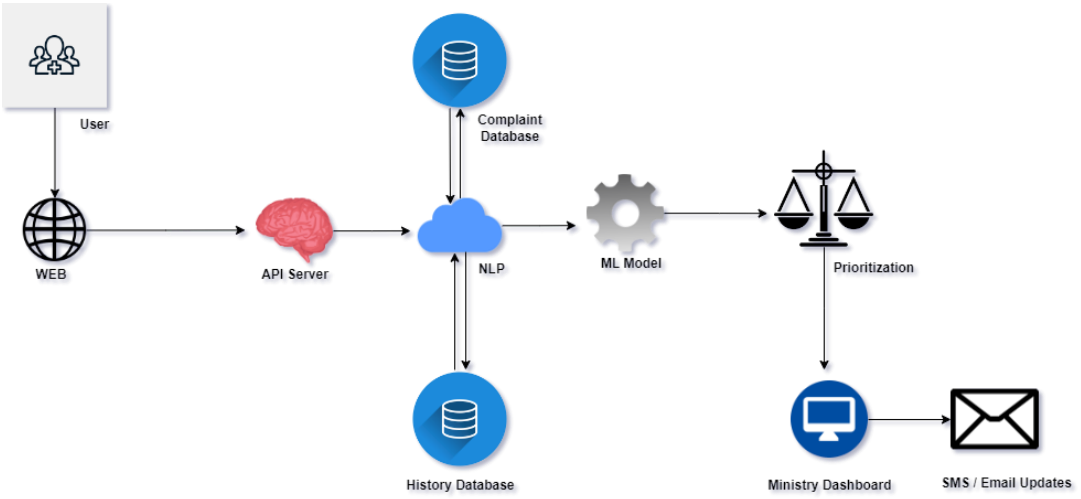

Figure 0.1:System Architecture Diagram

The digital system for handling complaints, GOV Connect LK, was built to handle citizen issues in urban and rural Sri Lanka. The platform supports English and code-mixed languages, to give access to all the citizens, residents to report on issues like hospital staff shortages, potholes, water supply, missing garbage collection, and issues with school infrastructure. It also links with Ministries such as health, education, highways, water supply, environment, public administration, transport, power and energy, agriculture and labor.

GOV Connect LK consists of two elements. First, a language processing engine that automatically analyzes the text of the complaint, without the need for an additional software application, and automatically categorizes it into categories (e.g., health, roads, water) and urgency (high, medium, low). The flagged complaints can thus be sent through to the relevant ministry. Second, a data-driven model which identifies past complaints and forecasts completion times and feasibility factors to determine how long the complaint could take and how feasible it is to do so again, helping to prioritize complaints that will have the most impact.

The mechanism also allows people to participate in the process by voting on the complaint posted by others and providing greater visibility to issues that many others are experiencing (they might even define the issue differently from each other). All submitted complaints are ranked by the ministry dashboards by urgency, time of expected resolution and number of votes.

This process allows transparency, faster resolution of problems, and better delivery of public services to the community and will give citizens a larger voice concerning government.

3.2 Component Specific System Diagram

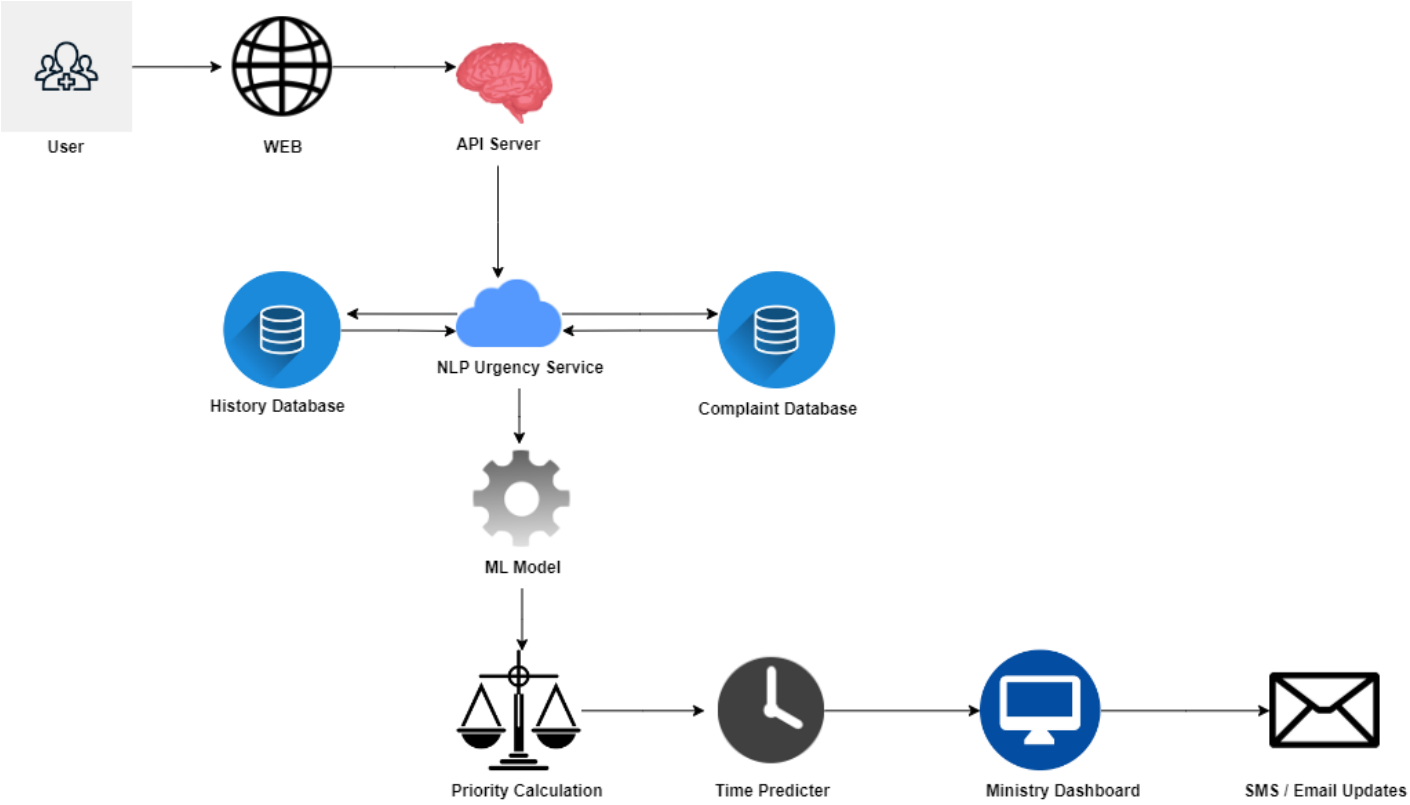

Figure 0.2:Component Architecture Diagram

The process is initiated by people submitting complaints through a user interface (presumably designed for accessibility) with informal Sinhala-English code-mixed textual input. These submissions contain basic information such as type of complaint, perceived urgency and government department to whom it is addressed, which again is the first input driving the system's operation.

AI Processing

Once submitted, the complaints are handled in the AI engine that is composed of two essential parts: The Natural Language Processing (NLP) module is used to analyze the text and categorize complaints into relevant categories and determine their urgency level, which was customized to deal with the linguistic diversity of Sri Lanka. At the same time, a Machine Learning (ML) regression algorithm such as XGBoost or Random Forest, trained on historical or simulated data, calculates the expected resolution time based on factors such as the type of the complaint, department and urgency, and provides a data-driven basis for prioritization.

Data Storage

Complaint information, urgency rating, estimated resolution time, and citizen voting information are stored in a centralized MongoDB database. This storage solution is scalable and provides quick querying facilities to allow the system to handle large amounts of data (e.g., 100,000+ records) efficiently and enable real time operations throughout the platform.

Priority Score Calculation

The backend system, built with Node.js and Express.js, computes a dynamic priority score for every complaint. The hybrid prioritization algorithm combines three important dimensions: urgency (which is calculated from an NLP analysis), estimated resolution time (calculated based on learning from the ML model) and voting signals from citizens (in the form of community votes). The score is calculated dynamically and updated in real time as

new data like more votes comes in to ensure a fair and responsive prioritization process.

Immediate Processing and Feedback

Integrated feedback loops ensure the system is continuously operated with real-time processing, tracked and adjusted on the fly as a response to changes in the status of complaints. As part of this review, alerts (e.g. email or SMS) are generated to notify citizens of updates or resolutions - providing enhanced transparency, accountability and continued engagement between citizens and government officials.

Officer Dashboard Engagement

Government officers communicate with the system through an interactive dashboard created using React.js, which is the center point of complaint management. This dashboard allows officers to sort and filter complaints by priority score, urgency, department, or type and to visually see what is happening in real-time (e.g., trends in the distribution of complaints). Complaint statuses can also be updated by officers, and the update then propagates back into the system for instant effect.

Notification and Closure

When the resolution or change of status occurs, the backend updates the database with the most recent information and the notification system sends out notifications to the appropriate citizens. This step closes the feedback loop, thus making sure that the public is aware of outcomes, thus strengthening trust and keeping communication channels open with government services.

3.3Project Requirements

3.3.1Functional Requirements

Function Name	Input	Processing	Output
Dataset Collection or Simulation	Complaint data (resolution time, department, urgency, type)	Collect or simulate dataset with required attributes	Structured dataset for model training
Regression Model for Resolution Time Prediction	Complaint features (type, department, urgency)	Apply ML regression model (e.g., XGBoost/Random Forest) to predict resolution time	Predicted resolution time with accuracy target ($R^2 > 0.85$)
Hybrid Prioritization Algorithm	NLP urgency score, predicted resolution time, citizen voting	Combine inputs to calculate dynamic priority score	Ranked list of complaints by priority
Backend System Development	Complaint records, API requests	Node.js/Express backend handles storage, computes scores, manages requests	Updated database, API responses
Interactive Officer Dashboard	Complaint and priority data from backend	Render React.js UI with sorting, filtering, charts, and update options	Officer dashboard with interactive analytics and complaint management
Feedback Loops and Notifications	Complaint status updates, citizen contact details	Track updates, trigger notifications via dashboard, email, or SMS	Citizens receive real-time updates, transparent complaint handling
API Integration	Regression model outputs, NLP outputs, priority scores	RESTful API connects backend, NLP system, and dashboard	Seamless data exchange across components

Table 2:Functional Requirements

3.3.2.Non-Functional Requirements

NFR	Testable NFR
Performance	system needs to be able to accept at least 100 complaint records/second with <2s response time.
Scalability	System must provide high scalability with up to 10,000+ active complaints and scale horizontally without performance degradation
Availability	System availability should be 99.5% or better during working hours. Reliability No more than 1 failure per 1000 transactions under normal load.
Reliability	No more than 1 failure per 1000 transactions under normal load.
Usability	Users should be able to perform the key functions in <3 clicks and the process of submitting a complaint should be done within <2 minutes.
Maintainability	System should be maintainable such that bug fixes or feature addition should be available within 24 hours without any major downtime.
Compatibility	Dashboard should work on modern browsers (Chrome, Firefox, Edge, Safari) without rendering problems..
Accuracy	The accuracy of an ML model should have a RMSE of less than 15% on that test data set.
Auditability	System should record 100% of the events related to the lifecycle of the complaint and allow authorized auditing of the events.

Table 3:Non Functional Requirements

3.4. Tools and Technologies

3.4.1. Tools

Development Environment

The main code editor to write, debug and manage code written in Node.js, Express, React.js and Python will be Visual Studio Code. It is compatible with JavaScript, Python and Git extensions, which help to develop and manage projects easily.

Machine Learning Tools

Python will be used as the primary language to develop the regression model, and Pandas and NumPy will be employed to create, clean and preprocess complaint attributes (resolution time, department, urgency and type). To achieve the desired results, XGBoost or Random Forest will be used to train the regression model in order to estimate the time of complaint resolution.

Backend Development

The runtime environment will be provided by Node.js and the RESTful APIs will be developed using the Express.js to combine regression model, NLP outputs, and officer dashboard and process the complaints, API requests, and priority score calculation.

Frontend Development

An interactive officer dashboard based on React.Js will be created, providing real-time updates, sorting, filtering and analytic visualizations so that officers can monitor and manage complaints effectively.

Database Management

The complaints, citizen votes and priority scores will be stored in the mongoDB, which is a NoSQL database to maintain scalability, quick queries and allow smooth interaction with the back-end system.

Version Control & Collaboration

A version control software will be GIT, which will be used to monitor the changes in the code, and the project repository will be hosted at the GitHub in order to support collaborative development between the team members.

Testing & Monitoring

RESTful APIs will be tested using Postman that will ensure a seamless connection between the regression model, NLP system, backend, and dashboard.

3.4.2 Technologies

The smart complaint management system hybrid prioritization component is based on the advanced technologies that allow predicting the time of the real resolving and hybrid prioritization as well as interactive officer dashboard. Such technologies provide the precise, scalable, and effective management of the complaints of people.

Machine Learning (XGBoost): XGBoost will be used to construct an effective regression model to predict the time required to resolve a complaint, achieving more than 85 percent accuracy (R 2). The model employs the following features to make quick and accurate predictions to control real-time decisions; complaint type, department and urgency.

Web Development (React.js and Node.js): The frontend officer dashboard is created to be based on React.js, which provides such interactive capabilities as real-time sorting, filtering, and analytic visualization of trends in complaints. Backend: API requests and priority score calculations are implemented with Node.js and Express to enable a smooth amount of integration with the regression model and dashboard.

NoSQL Database (MongoDB): MongoDB is implemented to store complaints, citizen votes and priority scores. It has a scalable architecture that can support high-volume data management (100,000+ records) and flexible querying in order to facilitate efficient real-time operations.

Cloud Computing (AWS): AWS services, such as EC2, which offers computation, and S3, which offers storage, offer a scaled deployment environment. This configuration is capable of managing 10,000 daily complaints and more than 1,000 concurrent API requests with low latency (less than 500ms), and therefore it can perform well when there is high traffic.

API Security (JWT & HTTPS): JSON Web Tokens (JWT) with HTTPS protocols ensure the safety of API endpoints and data transfer, protecting confidential information on complaints and control access to all parts of the system.

3.5.Implementation

The deployment of the hybrid prioritization component concerning the smart complaint management system is based on creating a model of prediction of the time of the resolutions, hybrid prioritization algorithm, and interactive officer dashboard. The system uses the latest technologies to provide the ability to make decisions in real-time, scale the system, and maintain transparency in managing complaints publicly. The implementation will be split into the front-end and back-end development, which coincides with the sub-objective and tasks.

Front End Development -

The front-end officer dashboard will be written in React.Js, which will form an interactive and accessible interface to real-time visualization of prioritized complaints and analytics. The react.js modular architecture allows building of reusable components, e.g. list of complaints, priority score display and visualization charts (e.g. complaints per department or department urgency patterns). React hooks and virtual Dom provide smooth, real time updates at the least possible latency so that officers can sort, filter and view the status of complaints in real time. D3.js is going to be incorporated to make the visualizations dynamic, i.e., bar charts or heatmaps, to use in the highlighting of the trends in complaints and their resolution. The dashboard will be designed to comply with the WCAG 2.1 guidelines towards accessibility that will allow usability by all the officers, such as those with disabilities, through accessibility features such as keyboard navigation and compatibility with screen readers. Status updates and trigger notifications (e.g. via email/SMS) will be shown as real-time feedback loops

to citizens, in-line with backend operations.

Back End Development -

Back end will be developed in Node.js and Express.js using powerful JavaScript framework to store the complaints, calculate the priority score, and respond to real time API requests. This will use express.js as the API layer which will combine the XGBoost regression model (trained in Python) to predict the complaint resolution time with an accuracy of over 85% (R 2) based on attributes such as complaint type, department and urgency. The backend will also be integrated to MongoDB which is a NoSQL database that will hold complaints, citizen votes and priority scores which will accommodate high volume data (100,000+ records) with quick querying capabilities. The hybrid prioritization algorithm shall be coded in Node.js and involve a combination of NLP-generated urgency scores, estimated resolution times and voter feedback as citizen votes into a weighted priority score and then provided as an API in 500ms to be used in real-time decision-making. RESTful APIs will allow a smooth interaction with the NLP element (urgency scores) and the front-end dashboard to have low-latency communication (less than 1s). The feedback loops will be responsible to update complaints statuses in MongoDB and send notification via external gateways (e.g. email/SMS), which will ensure transparency and user interaction.

Database Management -

The NoSQL database that will be utilized in managing the data of complaints, votes of citizens, priority scores, and system metadata of the hybrid prioritization component of the smart complaint management system will be MongoDB. Its plastic schema design can accommodate a wide range of data, including details about complaints (resolution time, department, urgency, type), voting history, and analytics history. The database will be hosted using AWS (e.g., deployed on Amazon DocumentDB with support of MongoDB or a self-managed instance of MongoDB) and will provide scalability and performance needed to process high volumes of complaints each day (10,000 or more complaints daily) and provide real-time querying in the officer dashboard.

The anonymized complaint data will be stored in MongoDB to assist in the training and optimization of the resolution time prediction model (XGBoost) to understand the relationship between the features of complaints and the results of the resolution over a period of time. The database will be efficient to query real-time dashboard updates, e.g., by priority score to sort complaints or to show analytics (e.g. department specific complaint trends), with query response times less than 500ms. It will also capture systems performance measures, e.g., prediction accuracy (where R 2 should be above 0.85) and API response times, to confirm that the system is effective and permit future deployment, e.g. adding other prioritization variables (e.g. social media signals). The scalability of MongoDB will support 1,000+ simultaneous users, which will provide effective performance of the public service delivery.

3.6.Deployment

The hybrid prioritization component will be developed for the smart complainant management system to deliver a scalable, reliable and accessible platform for prediction, prioritization and real-time interaction with the officer dashboard for complaints resolution. The solution is developed using modern cloud infrastructure and containerization to provide seamless integration between the regression model, backend processes and front-end visualization to enable efficient delivery of public services in Sri Lanka..

Deployment will consist of the following:

- **Model Hosting**:

The fine-tuned XGBoost regression for prediction of complaint resolution times will be deployed on AWS (e.g., SageMaker or EC2 with GPU support), such that inference time is low (<2s per prediction). The model will be trained using historical or simulated dataset on complaints and then offered to the backend through a RESTful API for real-time integration for priority score calculation.

- **Web Hosting**:

The app, which is made up of a React.js front-end and Node.js with Express.js back-end, will be containerized using Docker in order to maintain consistent functionality across environments. Being hosted on AWS EC2, the application will connect to MongoDB (using Amazon DocumentDB or a self-managed instance) to provide efficient data handling and real-time updates on the dashboards. The configuration will allow transparent presentation of the officer dashboard, where the complaints would be prioritized and analytics can be presented (trend of complaints, etc.).

- **Scalability & Access**:

AWS auto-scaling will allow the system to allow 10,000 daily complaints and 1,000+ simultaneous API requests with <500ms latency, providing high resiliency under high traffic load. The dashboard will be responsive across desktop and mobile browsers and in line with WCAG 2.1 accessibility standards to be accessible to all users including those with disabilities using assistive technology such as screen readers. Complaint statuses will be updated in real time with little lag and notifications (e.g. email/SMS) will be sent as required.

3.7. Evaluation of the system

The Hybrid Prioritization Component will be evaluated on the accuracy of the time prediction resolution model, performance of the hybrid prioritization algorithm, the backend system, the usability of the officer dashboard and the reliability of the feedback loops.

Resolution Time Estimate Accuracy:
The predicted resolution times and the actual or simulated resolution time will be compared with the regression model (XGBoost or Random Forest) fitted on the historical or simulated complaint data. The goal is to make the model accurate enough in its estimation to allow real-time decision making.

Hybrid Prioritisation Effectiveness:
By contrasting the hybrid priority score (the combination of urgency (from NLP), estimated time to resolution, and community voting) vs. the actual resolution times, we will also be able to analyze how well the hybrid priority score prioritizes high impact / urgent complaints vs. those that are less meaningful. The evaluation will

ensure that emergency issues are given priority above other issues but not at the cost of soliciting community input or seeking resolution of issues that can be relatively easily resolved.

Backend System Reliability:
The Node.js/Express based backend will be tested to ensure that they consistently store complaints, route API requests to their appropriate destination and produce priority scores. The evaluation will focus on response times and correct handling of information at the time of the complaint, prioritization and information retrieval.

Officer Dashboard Usability:
The officer dashboard will be tested for interactivity (sort, filter and view complaint analytics) using React.js Usability testing needs to be done to make sure that the dashboard can be easily navigated by officers, that priority scores can be easily understood and that data can be used to support decision making.

Feedback Loop Functionalities:
To test the feedback mechanism, the system will determine if the status of the complaints is accurately updated in real time and whether the notification is given to the citizens on time. This creates a window of transparency and continuous dialogue between citizens and government officials.

Overall, the evaluation confirms that the system meets its main objectives of making accurate predictions of resolutions, intelligently and fairly prioritizing cases, having a dependable backend, enabling users to interact with the dashboard, and giving transparent feedback to citizens.

4.WORK BREAKDOWN STRUCTURE

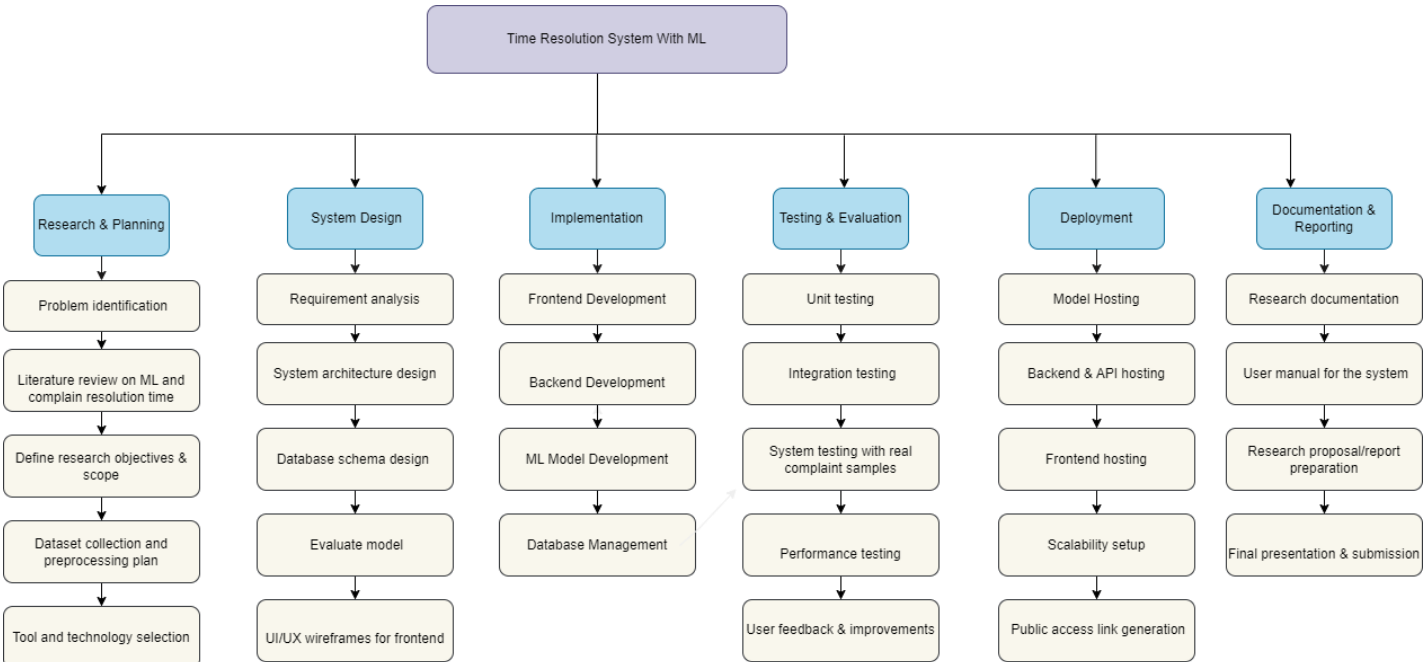

Figure 0.1:Work BreakDown Structure

The Time Resolution System development using Machine Learning (ML) has been divided into six major phases where each phase has certain and specified tasks to establish a systematic approach from the point of planning to the implementation and reporting of work.

- Research and Planning

The project starts with the identification of the problem and literature review of machine learning approaches for complaint resolution time. The activities in this phase include clarifying research objectives and scope, planning for data collection and preprocessing, and choosing appropriate tools and technologies. These activities form the basis for the technical and methodological orientation of the system.

- System Design

After the planning phase, the requirement analysis for the planning stage and the basic system architecture design takes place in the system design phase. A database schema which could be used to process complaint records efficiently is designed and initial evaluation of the selected model is carried out. Also, UI/UX wireframes are developed for the officer dashboard so that the system is easy to use and meets usability objectives.

- Implementation

The implementation stage is the real system development. The frontend is implemented with React.js to offer an interactive officer dashboard. The backend is created with Node.js and Express.js to handle complaints, API requests, and priority scores. The ML model is constructed and trained (such as XGBoost/Random Forest) as a predictor of resolution times and the database management (MongoDB) is established to store complaint, voting,

and analytics records.

- Testing and Evaluation

In this phase, extensive testing is conducted to assure the reliability and effectiveness of the system. Unit testing checks components of the system, and integration testing checks that frontend, backend, database and ML model communicate with one another. The system is tested using actual or simulated complaint samples for its performance validation. Further performance testing looks at system speed and responsiveness, and user feedback is gathered for system improvements.

- Deployment

The system is made available to users in the deployment phase. ML model is hosted on the cloud infrastructure and back end APIs and frontend dashboard is deployed which allows real time usage. Scalability patterns are used to process huge numbers of complaints and simultaneous requests. Finally, a public access link is created which helps the officers and citizens to interact with the system smoothly.

- Recording and Reporting

The last phase is documentation and reporting. Documentation of the research is prepared that will outline methods and results, and a user manual is developed that will outline how officers and administrators can use the system effectively. Research proposal and final report are developed, the project is finalized with the preparation and delivery of the final presentation and submission.

5.GANTT CHART

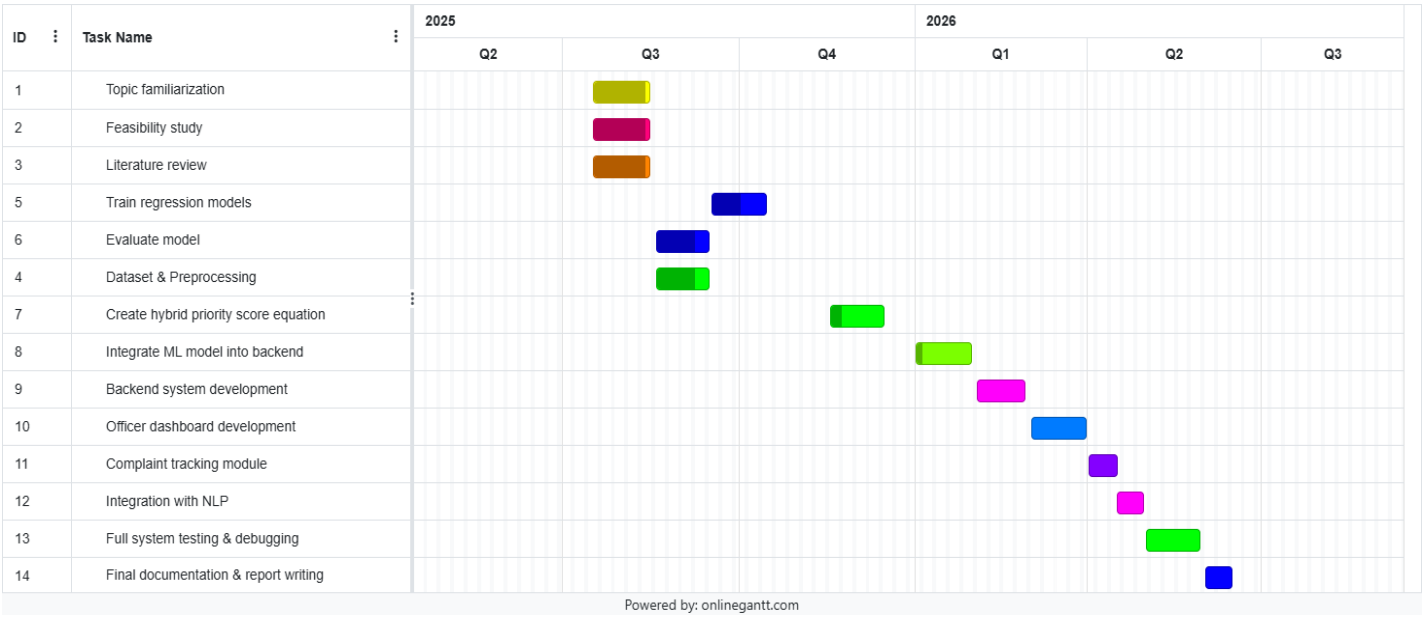

Figure 0.1:Gannt Chart

Milestones

- Proposal Submission – Aug 2025
- Progress Presentation 1 – Dec 2025
- System Prototype – Mar 2026
- Final Presentation – Jun 2026

The Gantt chart provides the project activities including starting and ending dates, duration in days and key milestones. This hierarchical timeline to project activities ensures that tasks can be performed consistently and the project team can keep track of their progress.

In June 2025, the project start with topic familiarization (15 days) and then moving into a feasibility study (30 days) and the project team will likely be in this phase from June 2025 until the middle of July 2025. The project team will then complete a detailed literature review which should take 31 days, and the Project Team is expected to complete that milestone which fusion stage will complete in August of 2025, to the next key milestone: Proposal Submission:

The next stage will involve dataset manipulation and cleaning activities (31 days) during August-September 2025, the training of the Regression Models (30 days) September to October, and keep the model evaluation (31 days) activities moving through October-November. Then the project activities could propose the completion of the hybrid priority score equation to complete the Fusion Stage 2 activities (30 days) during the moving into November-December 2025 completion, a project team milestone Progress Presentation 1.

Moving into January 2026 the project determination of implementation will cover (5 days) starting with NHLML integration and backend work, through to discussing the backend systems development effort (28 days of the project takes place in February 2026) with March (31 days) activities focused on the officer dashboard and once completed deliver the System Prototype (project milestone).
Following the completion of the prior activities with the System Delivery, the last bucket of work is wrapping up complain tracking and problem analysis implementation during April 2026 (15 days), transition of the NLP component from April to 2026 (15 days), system testing and debugging in May 2026 (31 days), and finish with final documentation and report writing in June 2026 (15 days) leading to the Final publication milestone.

6.COMMERCIALIZATION OF THE PRODUCT

The hybrid prioritization system on complainant management can be commercialized by focusing on government agencies, municipal councils and public service organizations that deal with the complaints of citizens. The product can be provided as a Software-as-a-Service (SaaS) solution that can be deployed on a multi-department basis with minimal IT infrastructure footprint.

Partnerships will be formed with national digital governance and e-governance initiatives to improve adoption of the system, making it a standard and shared tool to improve transparency, accountability and efficiency of public service delivery. Pilot projects can be run in a few districts to show their effectiveness, and the results of these should be able to produce case studies and numbers to encourage wider government take-up.

Moreover, a subscription-based model can be put into practice for international municipalities and NGOs dedicated to civic participation, where the hybrid prioritization algorithm can be tailored to local regulation, language and types of complaints. The solution can also be integrated with existing public service platforms to give dashboards, analytics, and real-time decision support for complaint officers.

7.BUDGET AND BUDGET JUSTIFICATION

Project Overview: The Hybrid Prioritization Component of GovConnect is built to be cost efficient, utilizing primarily free and open-source tools including Python, XGBoost, Node.js, React.js, MongoDB, and GitHub. The project only spends on basic things such as dataset creation, cloud hosting, and evaluation.

Dataset Collection & Simulation: The biggest budget allocation (25%) is for creating or simulating complaint datasets, which include information such as the type of complaint, urgent or not, department, and how long it takes to resolve it. This is costly but essential to the construction of an accurate prediction system.

Cloud Computing & Hosting: 20% of the budget is allocated to cloud servers, GPU-supported model training, and domain registration, ensuring the system's real-time performance and scalability to cater to many users.

Database & Storage: 15% of budget is spent on scaling the MongoDB database for efficient storage of complaint records, votes and analytics logs.

Development Tools: 15% is allocated for development tools, which includes low-end items such as paid plugins or UI libraries, while most of the work uses free tools such as VS Code, Postman and GitHub.

Testing and Evaluation: 10% of the budget is set aside for testing. This includes usability testing, browser and device compatibility testing, and performance benchmarking to ensure the system works correctly.

Internet & Miscellaneous Costs: Internet costs need 5% so that we can have a stable connection, and 10% to cover unexpecteds costs, such as new libraries or cloud resources being added.

Conclusion: Given the budget of 100,000 LKR as total budget, this allocation is feasible and adequate for developing, testing and deploying the hybrid prioritization system in order to offer research value and application benefits for public complaint management in Sri Lanka.

Category	Estimated Cost (LKR)
Dataset Collection & Simulation	25,000
Internet Cost	5,000
Cloud Computing & Hosting	20,000
Development Tools	10,000
Database & Storage	15,000
Testing & Evaluation Tools	10,000
Miscellaneous & Contingency	10,000
Total	100,000

Table 4:Budget

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